

AP Calculus AB: The Complete Exam Guide

All 8 units explained, the calculator and no-calculator split, a free-response playbook with justification language, and 10 hard practice questions.

Hi, I'm **Pigo!** I'll drop tips all through this guide.

**3:10**

hours total
100 + 90 minutes

42+6

multiple-choice +
free-response questions

2/3

of the exam is
no-calculator

64–65%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two sections, four parts

Your AP score (1–5) combines multiple choice and free response, each worth half. Both sections are split into a no-calculator part and a calculator part, and most of the exam is done without one.

35%

MC Part A • 29 Q • no calc

15%

MC Part B • 13 Q • ...

17%

FR Part A • calc

33%

FR Part B • 4 Q • no calc

Exam structure

Section	Weight & timing	What you do
I • Multiple choice	50% of score 1 hr 40 min 42 questions	Part A: 29 questions, no calculator (35% of score). Part B: 13 questions, graphing calculator required for some (15%). Functions appear as equations, graphs, tables and words.
II • Free response	50% of score 1 hr 30 min 6 questions	Part A: 2 questions, graphing calculator required (about 17%). Part B: 4 questions, no calculator (about 33%). At least two questions use a real-world context.

HYBRID DIGITAL

Click in Bluebook, write on paper

You answer multiple-choice questions and read the free-response questions in the **Bluebook** app, then handwrite your free-response answers in a paper booklet. Practice writing clear, organized work by hand.

CALCULATOR RULES

Know your four calculator skills

On calculator parts you are expected to **graph** a function, **solve** an equation, find a **numerical derivative** and evaluate a **definite integral**. Check that your model is on College Board's approved list.

How to use this guide

Just starting?

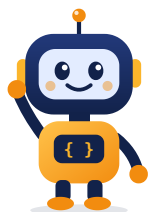
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 25 minutes, then use the final-four-weeks plan.



Pigo's tip: Every free-response question is scored on a 9-point rubric. Points come from setting up correctly and justifying, not just the final number.

SECTION I

02 The multiple-choice section

Part A gives you 29 questions in 62 minutes without a calculator (about 2 minutes each). Part B gives 13 questions in 38 minutes with a calculator (about 3 minutes each).

29

no-calculator questions
in Part A

13

calculator-allowed
questions in Part B

~2 min

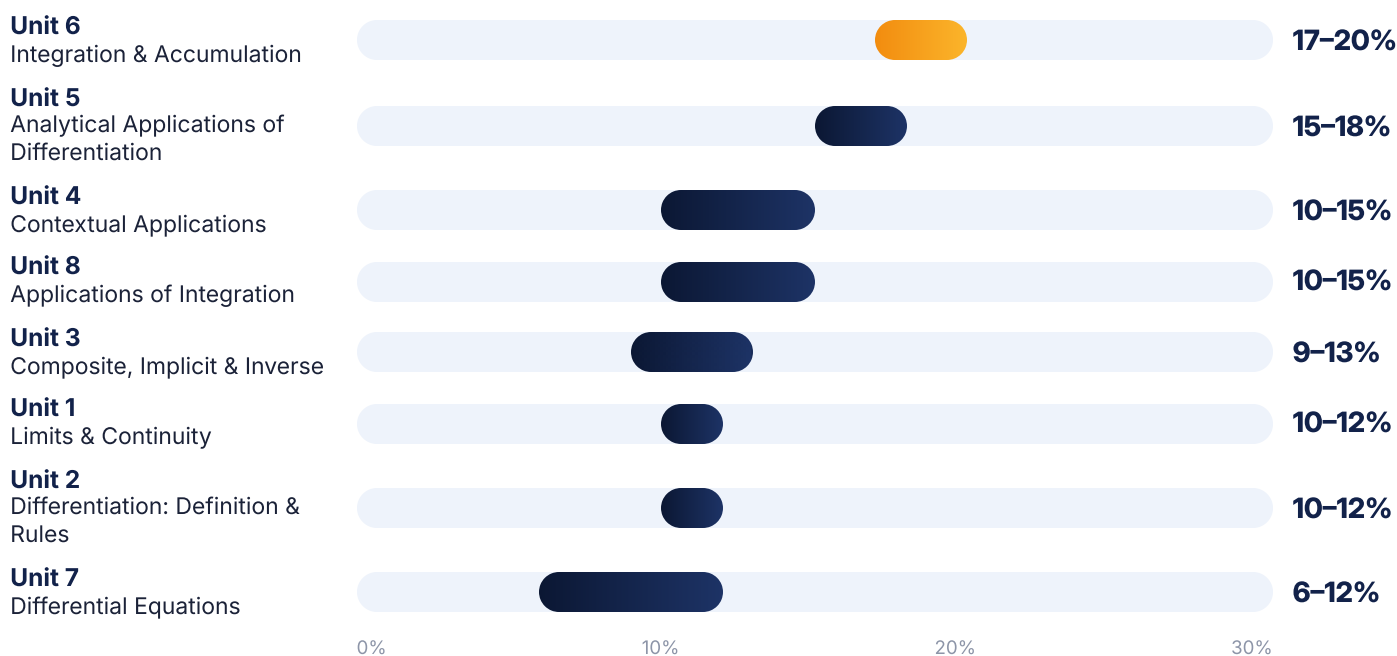
per question in Part A

0

points lost for a wrong
answer, so never leave
one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the exam. Bars show minimum to maximum.



What this means for your study time: Units 5 and 6, analyzing functions with derivatives and the meaning of the integral, are the biggest. They also drive most free-response questions.

Strategies that work on this section

- **Read graphs of f' carefully.** Many questions show the derivative, not the function. Increasing f means $f' > 0$, not f' increasing.
- **Know your limits toolkit:** factoring, conjugates, the special trig limits and L'Hospital's Rule.
- **Units tell you the answer type.** A rate in gallons per hour, integrated over hours, gives gallons.
- **In Part B, use the calculator** for definite integrals and solving. Store intermediate values, don't round.
- **Eliminate with estimates.** A quick sketch often rules out two choices.
- **Flag and move on** after 3 minutes. Come back once the quick points are banked.

03 WHAT THE COURSE COVERS

The 8 units, in plain English

Everything on the exam fits into these eight units.

UNIT 1 10–12%	Limits & Continuity What a function approaches, and where it breaks. Be ready to: evaluate limits from graphs, tables and algebra; use the squeeze theorem; find asymptotes; apply the definition of continuity and the IVT.	UNIT 2 10–12%	Differentiation: Definition & Rules The derivative as a limit and an instantaneous rate. Be ready to: use the limit definition; apply power, product, quotient rules and trig/exponential derivatives; connect differentiability and continuity.
UNIT 3 9–13%	Composite, Implicit & Inverse Harder derivative rules. Be ready to: apply the chain rule; differentiate implicitly; find derivatives of inverse functions and inverse trig functions; find second derivatives.	UNIT 4 10–15%	Contextual Applications Derivatives describing real change. Be ready to: interpret derivatives with units; solve motion and related-rates problems; use local linearization; apply L'Hospital's Rule.
UNIT 5 15–18%	Analytical Applications Using derivatives to analyze a function's shape. Be ready to: apply the MVT and EVT; find intervals of increase and concavity; justify extrema with first and second derivative tests; solve optimization problems.	UNIT 6 17–20%	Integration & Accumulation The integral as accumulated change. Be ready to: compute Riemann and trapezoidal sums; apply both parts of the FTC; find antiderivatives; use u-substitution; interpret accumulation functions.
UNIT 7 6–12%	Differential Equations Equations that describe how something changes. Be ready to: verify solutions; sketch and read slope fields; solve separable equations with initial conditions; model exponential growth.	UNIT 8 10–15%	Applications of Integration Average value, motion, area and volume. Be ready to: find average value; use integrals for position and total distance; find areas between curves; find volumes by cross sections and discs/washers.

AB or BC? AB covers Units 1–8. BC adds Units 9–10 (parametric, polar and series) and some extra topics. An AB subscore is also reported to BC students.

04 SECTION II The 6 free-response questions

90 minutes, 6 questions, 9 points each. Part A (2 questions, calculator) gets 30 minutes; Part B (4 questions, no calculator) gets 60 minutes.

Q1 · calc	Q2 · calc	Q3	Q4	Q5	Q6
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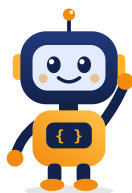
6 questions × 9 points = 54 raw points, worth 50% of your AP score

Rate Rates in and out Often Part A Water, people or cars enter and leave. You integrate rates for amounts, compare rates to decide increasing or decreasing, and find maximums. Amount at time t = starting amount + $\int$ (in – out).	Table Data in a table Riemann sums, MVT, IVT Estimate integrals with left, right, midpoint or trapezoidal sums, interpret them with units, and apply MVT or IVT to the data. Check the hypotheses aloud: "f is differentiable, so..."	Graph Graph of f' Analyze the hidden f Given the graph of a derivative, find extrema, concavity, inflection points and values of f using areas. Justify with f', never with "the graph goes up".
Motion Particle motion Position, velocity, acceleration Decide direction, speeding up or slowing down, displacement versus total distance, and position at a given time. Speeding up: v and a have the same sign.	DE Differential equations Slope fields + separation Sketch a slope field, use tangent-line approximation, and solve a separable equation with an initial condition. Separate, integrate, add +C, use the initial condition, then solve for y.	Area Area & volume Or implicit / analytic Area between curves and volumes by discs, washers or cross sections. Sometimes replaced by an implicit-differentiation question. Write the integral setup clearly; it earns points even without evaluating.

• A time plan for the 90 minutes

Part A Q1 15 min	Part A Q2 15 min	Q3 15 min	Q4 15 min	Q5 15 min	Q6 15 min
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About 15 minutes per question. In Part B you may return to Part A questions, but without a calculator.



Write the **setup** before you compute. An integral written correctly with limits earns points even if you run out of time.

FREE-RESPONSE PLAYBOOK

05

Anatomy of a full-credit answer

Justification is where most students lose points. Here is a short practice prompt with a full-credit answer.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (Part B style): The derivative of a function f is $f'(x) = (x - 3)e^{-x}$. (a) Find the x -coordinate of each relative extremum of f and classify it. Justify your answer. (b) On what interval is the graph of f concave up? Justify your answer.

Full-credit response

- ① (a) $f'(x) = 0$ when $x = 3$, because $e^{-x} > 0$ for all x .
- ② $f'(x)$ changes from negative to positive at $x = 3$.
- ③ Therefore f has a **relative minimum at $x = 3$** .
- ④ (b) $f''(x) = e^{-x} - (x - 3)e^{-x} = (4 - x)e^{-x}$.
- ⑤ $f''(x) > 0$ when $x < 4$, so f is **concave up on $(-\infty, 4)$** because $f'' > 0$ there.

- ① **Find the candidate** and say why the other factor can't be zero.
- ② **Name the sign change** of f' , the evidence the rubric wants.
- ③ **State the conclusion** in words: relative minimum, at $x = 3$.
- ④ **Show the derivative work.** The product rule earns its own point.
- ⑤ **Connect sign to conclusion** using f'' , not "the graph curves up".

Five habits for every free-response question

1

Label every part: (a), (b), (c).

2

Justify with f' or f'' signs, not words like "the graph".

3

Include units in every contextual answer.

4

Give decimals to 3 places on calculator parts.

5

Stuck? Write the setup integral or equation. It still scores.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

Forgot the **+C**? I'll take that point!



01 Justifying with "the graph"

Readers need a reason based on f' or f'' : " f' changes from $+$ to $-$ ", not "the graph peaks".

02 Displacement vs distance

Total distance is $\int |v(t)| dt$. Displacement is $\int v(t) dt$. They differ whenever the particle turns around.

03 Forgetting the chain rule

$d/dx[\sin(x^2)] = 2x \cos(x^2)$. The inner derivative is the most common missing piece.

04 Rounding too early

Store calculator values and round only the final answer to 3 decimal places.

05 Dropping **+C** or the initial condition

In separable equations, add $+C$ right after integrating and solve for it before isolating y .

06 Mixing up f , f' and f''

If the graph shown is f' , its zeros are f 's critical points and its slopes give f 's concavity.

07 Using a theorem without its conditions

MVT needs continuous on $[a, b]$ and differentiable on (a, b) . Say so when you use it.

08 Wrong limits after u -substitution

Either change the limits to u -values or switch back to x before evaluating.

09 Missing units in context

A rate in liters per minute integrated over minutes gives liters. Write it.

10 Leaving a part blank

Each part is scored separately. A correct setup can earn points even if the next step fails.

Quick self-check: if $f'(2) = 0$ and $f''(2) = -3$, what can you conclude about f at $x = 2$? (Answer: a relative maximum, by the second derivative test.)

07

PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, but keep reviewing old units as you learn new ones.



Sep – Oct
Foundations



Units 1–2: limits and derivatives

Master limit techniques and every derivative rule. Do 10 derivative drills a day until you rarely make sign errors.

Nov – Dec
Applications



Units 3–5 big MCQ share

Chain rule, implicit, related rates, MVT, extrema and optimization. Start writing justifications in full sentences.

Jan – Mar
Integrals



Units 6–8

Riemann sums, FTC, u-substitution, differential equations, area and volume. Do one released free-response question per week.

April
Full practice



Timed sections

Alternate full multiple-choice sections with full free-response sections under real timing, splitting calculator and no-calculator parts.

Early May
Sharpen



Justification polish

Rewrite your weakest justifications until they match rubric language. Review the traps list.

Exam day
May



AP Calculus AB exam

Sleep well, pack your approved graphing calculator with fresh batteries, and use the checklist at the end.

◆ The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: one full no-calculator Part A and two free-response questions	List your three weakest units
3 weeks out	Drill weak units; one free-response question of each type	80%+ on unit drills
2 weeks out	Full 3-hour practice exam	Finish every part on time
Final week	Light review: derivative/integral rules, justification phrases	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. No calculator unless marked. Give yourself **25 minutes** for all ten, then check page 10.

1 UNIT 1 • LIMITS

What is the value of the limit?

$$\lim_{x \rightarrow 0} (1 - \cos 3x) / x^2$$

- A 0 B $3/2$
C $9/2$ D 9

2 UNIT 2 • DERIVATIVE RULES

If $f(x) = x \cdot e^{2x}$, what is $f''(0)$?

- A 2 B 3
C 4 D 8

3 UNIT 3 • IMPLICIT DIFFERENTIATION

For the curve $x^2 + xy + y^3 = 3$, what is dy/dx at the point (1, 1)?

- A $-3/4$ B -1
C $3/4$ D $-1/2$

4 UNIT 4 • RELATED RATES

Water drains from an inverted cone of radius 3 ft and height 6 ft at $2\pi \text{ ft}^3/\text{min}$. How fast is the water level falling when the water is 4 ft deep?

- A $1/8 \text{ ft/min}$ B $1/4 \text{ ft/min}$
C $1/2 \text{ ft/min}$ D 2 ft/min

5 UNIT 6 • TRAPEZOIDAL SUM

Using a trapezoidal sum with the three subintervals shown, what is the approximation of $\int_0^6 f(x) dx$?

x	0	2	5	6
f(x)	4	8	3	5

- A 30 B 32.5
C 34 D 36

6 UNIT 6 • FTC

If $g(x) = \int_1^{x^2} \sqrt{1+t^3} dt$, what is $g'(1)$?

- A $\sqrt{2}$ B 2
C $2\sqrt{2}$ D 4

7 UNIT 5 • EXTREMA

The derivative of f is $f'(x) = (x-2)^2(x+1)(x-4)$. At which value(s) of x does f have a **relative minimum**?

- A $x = -1$ only
B $x = 2$ and $x = 4$
C $x = -1$, $x = 2$ and $x = 4$
D $x = 4$ only

8 UNIT 8 • MOTION

A particle moves along a line with velocity $v(t) = t^2 - 4t + 3$ for $0 \leq t \leq 4$. What is the **total distance** it travels?

- A $4/3$ B $8/3$
C 4 D $16/3$

9 UNIT 7 • SEPARABLE DE

$y = f(x)$ is the solution to $dy/dx = xy^2$ with $f(0) = 1$. What is $f(1)$?

- A $1/2$ B $2/3$
C $\sqrt{e}$ D 2

10 UNIT 8 • VOLUME

The region bounded by $y = \sqrt{x}$ and $y = x$ is revolved about the x -axis. What is the volume of the solid?

- A $\pi/12$ B $\pi/6$
C $\pi/3$ D $\pi/2$

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	C • 9/2	Use $1 - \cos u \approx u^2/2$ for small u (or L'Hospital twice): $(1 - \cos 3x)/x^2 \rightarrow (9x^2/2)/x^2 = 9/2$.
2	C • 4	$f'(x) = e^{2x}(1 + 2x)$. $f''(x) = e^{2x}(4 + 4x)$, so $f''(0) = 4$. Choice A forgets the product rule on the second derivative.
3	A • -3/4	$2x + y + xy' + 3y^2y' = 0$. At $(1, 1)$: $3 + 4y' = 0$, so $y' = -3/4$.
4	C • 1/2 ft/min	Similar triangles give $r = h/2$, so $V = \pi h^3/12$ and $dV/dt = (\pi h^2/4)(dh/dt)$. $-2\pi = 4\pi(dh/dt)$, so the level falls at 1/2 ft/min.
5	B • 32.5	Unequal widths: $2 \cdot (4 + 8)/2 + 3 \cdot (8 + 3)/2 + 1 \cdot (3 + 5)/2 = 12 + 16.5 + 4 = 32.5$.
6	C • $2\sqrt{2}$	By the FTC and chain rule, $g'(x) = \sqrt{1 + x^6} \cdot 2x$. At $x = 1$: $2\sqrt{2}$.
7	D • $x = 4$ only	f' changes sign only at -1 (+ to -, a maximum) and 4 (- to +, a minimum). $(x - 2)^2$ never changes sign, so $x = 2$ is not an extremum.
8	C • 4	$v = (t - 1)(t - 3)$ changes sign at 1 and 3. Distances: $4/3 + 4/3 + 4/3 = 4$. Choice A is the displacement.
9	D • 2	Separate: $-1/y = x^2/2 + C$. With $f(0) = 1$, $C = -1$, so $y = 2/(2 - x^2)$ and $f(1) = 2$.
10	B • $\pi/6$	Washers with outer radius $\sqrt{x}$ and inner radius x on $[0, 1]$: $\pi \int_0^1 (x - x^2) dx = \pi(1/2 - 1/3) = \pi/6$.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Drill derivative and integral rules (Units 2, 3 and 6) daily.

10 KNOW THE TARGET How students score

In recent years, about 64–65% of students earned a 3 or higher on AP Calculus AB, and about 20% earned a 5. Steady preparation is what moves a score up a full point.

64–65%

of students earned a 3 or higher in recent years

20%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, plus **two pencils and a dark-ink pen** for the free-response booklet
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ An **approved graphing calculator** with fresh batteries, in radian mode
- ☐ Plan your time: about **2 min** per Part A question and **15 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ Give calculator answers to **3 decimal places**

In the free-response section: label each part, show the setup before computing, justify with derivative signs or theorem conditions, and include units in every contextual answer.



ACT SAT GO • 1-ON-1 AP TUTORING

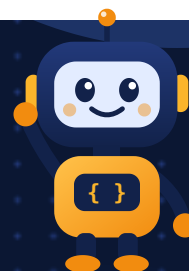
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Live 1-on-1 sessions on all eight units, with no-calculator drills after every lesson.

Free-response coaching

Practice every question type and get your justifications marked against rubric-style criteria.

Timed exam practice

Full-length multiple-choice sets with score reports, so you see progress before May.

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